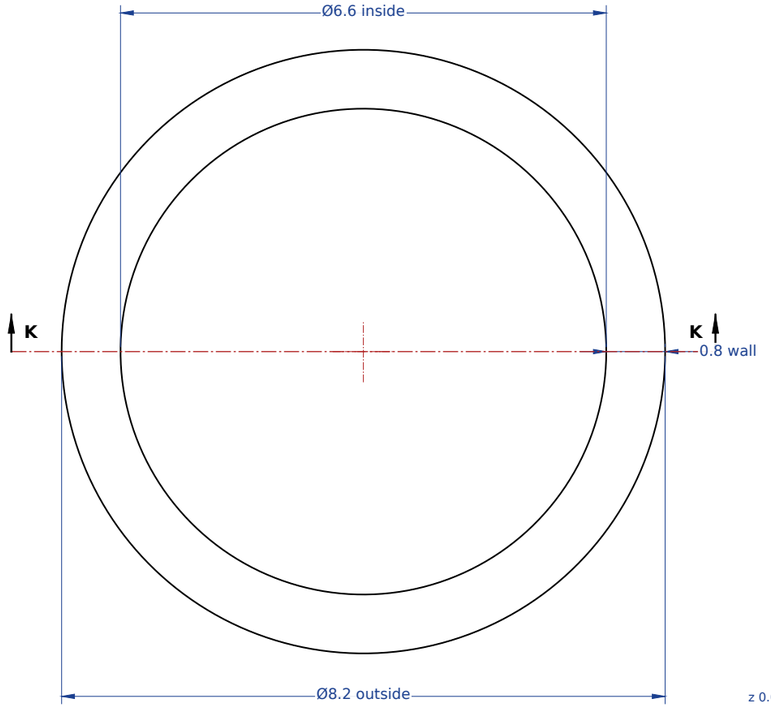
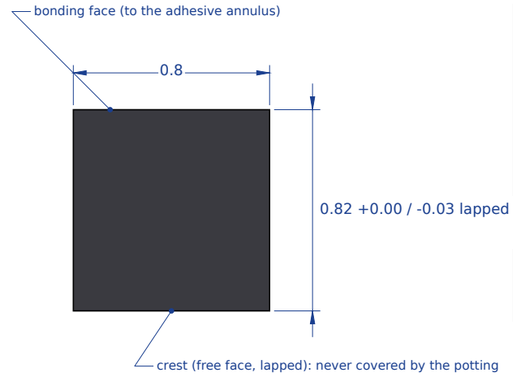
RING, seen on the face that bonds to the PCB 12:1



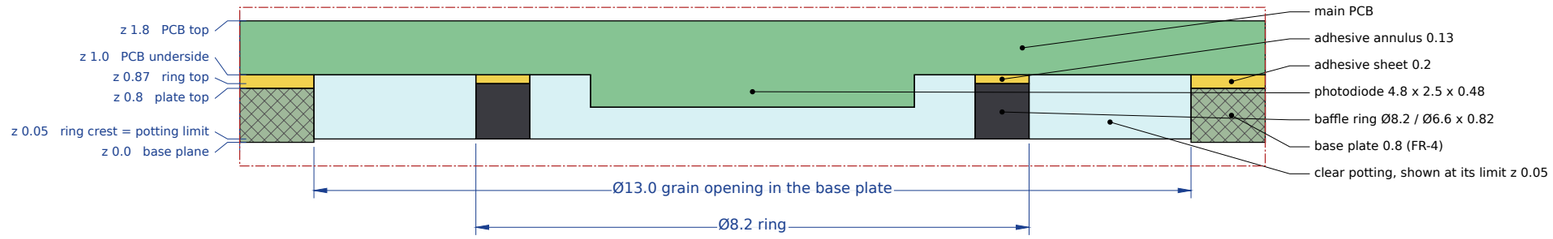
SECTION K-K (ring wall) 40:1



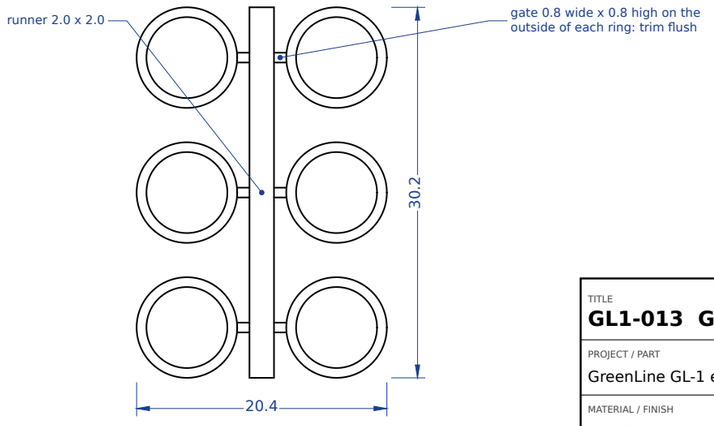
Clearance (model, check 2)	mm
photodiode corner to the ring bore	0.594
ring outside to the LED bodies (0402 on r 5.2)	0.6
LED outer corner to the opening wall	0.757
ring outside to the opening wall	2.4
potting over the photodiode / over an LED	0.47 / 0.5
ring crest above the base plane, as assembled (z)	0.05; range 0.00 to 0.13; accepted 0 to 0.15
potting surface (z), for the nominal crest	0.05 to 0.15, fill target 0.10

Item	Qty	Size	Material
Baffle ring	1	Ø8.2 / Ø6.6 x 0.82 lapped	MJF PA12, dyed black; printed 0.97 high
Adhesive annulus	1	Ø8.2 / Ø6.6 x 0.13	OPAQUE BLACK double-coated tape (dxf GL1-034)

AS ASSEMBLED - section y = -4.0 through the grain opening 11:1



PRINT FILE (baffle_ring_print): sprued set of 6 rings, each 0.97 high 2:1



NOTES

1. Light baffle between the grain LEDs and the photodiode of the FR-4 build. (The printed PA12 base plate GL1-012 carries the same ring itself, on three spokes.)
2. PRINTED 0.97 high as a sprued set (print file). WE cut the rings off, trim the gate flush and LAP the bonding face in a pocket gauge to the finished height 0.82 +0.00 / -0.03 for a base plate of 0.8 mm. For a plate lot of measured thickness t (over the mask) the finished height is t + 0.02: the crest then ends z 0.00 to 0.13 above the base plane and can never stand proud of it. (Lapped to 0.82 regardless of the plate it could end at z -0.03 to 0.22.) The crest is the as-printed face.
3. Wall 0.8 mm (the printing minimum). Order two sets.
4. ASSEMBLY, before the base plate is laminated: PCB component-side down in the lamination jig on its two sprung pins; annulus onto the ring; the ring tool, registered on the same two pins, places the ring round the photodiode (0.59 mm to the photodiode corners and to the LEDs). After lamination: crest between the base plane and 0.15 mm inside it (depth gauge). Then the opening is potted clear: surface from the crest to 0.1 inside it, never nearer the base plane than z 0.05 (README, steps 11 and 12).
5. Ring and annulus black and opaque at the LED wavelengths, low fluorescence: light-leak test (0.8 mm dyed MJF PA12; a clear tape would be a light slot between LEDs and photodiode).
6. PROVISIONAL - not released. Nothing has been built.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE

GL1-013 Grain baffle ring (FR-4 build)

PROJECT / PART		
GreenLine GL-1 electronic ball marker - grain baffle ring + adhesive annulus		
MATERIAL / FINISH	PROCESS	
PA12 (HP 3D HR PA 12), dyed black, not smoothed	MJF, sprued set; height lapped by us; annulus: black tape	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	12:1 / 40:1 / 11:1 / 2:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1 / D-EVT1	1 of 1 2026-11-11